Braid Arrangements for Concurrent Programs

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Abstract

Precubical sets are a standard tool in the study of concurrent programs. A recent paper by Paliga and Ziemiański showed that the fundamental group for the path space on the final precubical set was exactly is the Artin braid group. However, they also showed the induced braid subgroups for euclidean sets trivialized. In the world of hyperplane arrangements, this kind of trivialization can be resolved by working in the "complexified" space. We use this same approach, to build a new, complexified version of the path space, Ch^*. This is homotopy equivalent to the Salvetti Complex of the braid arrangement in for the cube, but defined over any precubical set. By Salvetti's theorem, then, $\pi_1|\mathrm{Ch}^*(\square^n)|$ is is the pure braid group. This gives us a new braid invariant for concurrent programs.

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1 Introduction

This work is in 3 parts. First we'll recap the relevant material in brief. Second, we will perform the construction of the "complexified" path space. Third, we will study some of its properties.

To fix notation, let Cube be the box category: its objects are the cubes $\square^n$ for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on Cube; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set $K[n]$ of n-cells, together with face maps

$$d_i^\epsilon : K[n] \longrightarrow K[n-1] \qquad (1 \leq i \leq n,\ \epsilon \in \{0,1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write $\square^n$ for the standard n-cube, the presheaf represented by $\square^n \in \mathrm{Cube}$.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\mathrm{init}_K, \mathrm{fin}_K \in K[0]$, and morphisms are required to preserve them; write $\square^*_*$ for the resulting category. Each cube $\square^n$ is bipointed by its initial vertex $(0, \ldots, 0)$ and final vertex $(1, \ldots, 1)$.

2 Standard build

We recall three constructions on $\square^*_*$ from Paliga–Ziemiański.

The first is the **wedge** $\vee$. Given $P, Q \in \square^*_*$, the wedge $P \vee Q$ is the pushout gluing P's final vertex to Q's initial vertex:

$$\begin{tikzcd}
\square^0 \arrow[r, "\mathrm{init}_Q"] \arrow[d, "\mathrm{fin}_P"'] & Q \arrow[d] \\
P \arrow[r] & P \vee Q \arrow[ul, phantom, "\ulcorner", very near start]
\end{tikzcd}$$

The wedge $P \vee Q$ is bipointed by init_P and fin_Q, and $(\square^*_*, \vee, \square^0)$ is monoidal with unit the point $\square^0$. The **serial wedge** of a list of naturals is

$$\bigvee[a_1, \ldots, a_n] := \square^{a_1} \vee \cdots \vee \square^{a_n},$$

with $\bigvee[\,] = \square^0$.

The second construction is the category of **cube chains**. For $K \in \square^*_*$, let $\mathrm{Ch}(K)$ be the full subcategory of the slice $\square^*_*/K$ spanned by the serial wedges: an object is a bipointed map $\bigvee a \to K$, and a morphism $(\bigvee a \to K) \to (\bigvee b \to K)$ is a bipointed map $\bigvee a \to \bigvee b$ commuting over K. Such a morphism runs from a finer chain to a coarser one; for instance the two-edge path $\bigvee[1,1] \to \square^2$ carries a morphism to the identity chain $\square^2 \xrightarrow{\mathrm{id}} \square^2$.

The third construction is **refinement**. A *chain* in K is a finite sequence $[c_1, \ldots, c_l]$ of cells of K, each of dimension at least 1, such that $\mathrm{fin}(c_i) = \mathrm{init}(c_{i+1})$ for all i, with $\mathrm{init}(c_1) = \mathrm{init}_K$ and $\mathrm{fin}(c_l) = \mathrm{fin}_K$; here $\mathrm{init}(c)$ and $\mathrm{fin}(c)$ denote the images under $c : \square^{\dim c} \to K$ of the initial and final vertices of the cube. A *refinement* replaces a cell c_i by a chain of the cube $\square^{\dim c_i}$, post-composed with c_i. Let $\mathrm{Refine}(K)$ be the category of chains, with morphisms generated by refinements and oriented as in $\mathrm{Ch}(K)$ (finer to coarser).

Finally, two properties of K from the Configuration Spaces paper recur. We say K *admits altitude* if there is a function $\mathrm{alt} : K[0] \to \mathbb{Z}$ with $\mathrm{alt}(\mathrm{fin}(e)) = \mathrm{alt}(\mathrm{init}(e)) + 1$ for every edge $e \in K[1]$. We say K is *non-self-linked* (NSL) if every map $\square^n \to K$ is mono.

A few things to note

Lemma 2.1. *The $\vee$ operator is a monoidal product on $\square^*_*$.*

Lemma 2.2. $\mathrm{Ch} : \square^*_* \Rightarrow Cat$ *is a lax monoidal functor from $(\square^*_*, \vee)$ to $(Cat, \times)$.*

Lemma 2.3. *If K admits altitude and is NSL, then* $\mathrm{Ch}(K)$ *and* $\mathrm{Refine}(K)$ *are equivalent, so* Ch *is strong monoidal there.*

Lemma 2.4. *Serial wedges admit altitudes, and are NSL*

This is all the foundational material we need.

3 Wedge Machinery

In an effort to keep our proofs better factored, we establish some results here that let us extend construction on Cube to constructions on $\square_*^*$.

Lemma 3.1. *Any functor* $P : \mathrm{Cube} \Rightarrow Cat$ *extends, by left Kan extension along* $\mathrm{Cube} \hookrightarrow \square$*, to a functor* $E(P) : \square_*^* \Rightarrow Cat$*. If* $P(\square^0) = \emptyset$*, then* $E(P)$ *is strong monoidal*

$$E(P) : (\square_*^*, \vee, \square^0) \Rightarrow (Cat, \sqcup, \emptyset),$$

and for every serial wedge

$$E(P)(\bigvee a) \;\simeq\; \coprod_i P(\square^{a_i}).$$

Proof. As a pointwise left Kan extension (a coend over Cube), $E(P)$ is cocontinuous, so it sends the wedge pushout $A \leftarrow \square^0 \to B$ to $E(P)(A \vee B) = E(P)(A) \sqcup_{P(\square^0)} E(P)(B)$. When $P(\square^0) = \emptyset$ this pushout is the coproduct, so the comparison map is an isomorphism; the formula for $\bigvee a$ follows by induction on the number of summands. $\square$

Lemma 3.2. *Any functor* $P : \mathrm{Cube}^{op} \Rightarrow Cat$ *extends, by right Kan extension along* $\mathrm{Cube} \hookrightarrow \square$*, to a* Cat*-valued presheaf* $E(P) : \square_*^{*op} \Rightarrow Cat$*. If* $P(\square^0)$ *is terminal, then* $E(P)$ *is strong monoidal*

$$E(P) : (\square_*^{*op}, \vee, \square^0) \Rightarrow (Cat, \times, \mathbf{1}),$$

and for every serial wedge

$$E(P)(\bigvee a) \;\simeq\; \prod_i P(\square^{a_i}).$$

Proof. Dually, $E(P)$ is a pointwise right Kan extension, hence continuous, and being contravariant it sends the wedge pushout to the pullback $E(P)(A \vee B) = E(P)(A) \times_{P(\square^0)} E(P)(B)$. When $P(\square^0)$ is terminal this pullback is the product, so the comparison is an isomorphism; the formula for $\bigvee a$ follows by induction. $\square$

Together, these give us tools for lifting and decomposing cube maps to tensor maps.

We now instantiate this machinery. The first case is the coordinate functor.

Definition 3.3. Let $\mathrm{Coord} : \mathrm{Cube} \Rightarrow \mathit{Set}$ send the cube $\square^n$ to its set of directions $[n] := \{1, \ldots, n\}$, and send a face embedding $f : \square^n \to \square^m$ to the monotone injection $[n] \hookrightarrow [m]$ that names the n directions f leaves free. Note $\mathrm{Coord}(\square^0) = [0] = \emptyset$, so the first lifting lemma applies to Coord: the extension $E(\mathrm{Coord})$ is strong monoidal, with

$$E(\mathrm{Coord})(\bigvee a) \;\simeq\; \coprod_i [a_i].$$

For a morphism $f : \bigvee a \to \bigvee b$ we write $coordMap(f) := E(\mathrm{Coord})(f)$ for the induced function on coordinate sets,

$$coordMap(f) : \coprod_i [a_i] \longrightarrow \coprod_j [b_j],$$

which records where each coordinate of $\bigvee a$ is sent. We regard the coordinate set $\coprod_i [a_i]$ as linearly ordered by the ordinal sum $[a_1] + \cdots + [a_n]$ — first by summand, then by direction within a summand — and it is this order that we use below to speak of the inversions of $coordMap(f)$.

Lemma 3.4. *For any $f : \bigvee a \longrightarrow \square^m$ of morphisms in $\square$, the function $coordMap(f)$ is injective.*

Proof. We recurse on a. If a is empty, then $coordMap(f) : [0] \to [m]$ which is always injective. For the induction case, we have $f : (\bigvee a) \vee \square^n \longrightarrow \square^m$. Now, by the universal property of the pushout, this splits into $h : \square^n \to \square^m$ and $g : \bigvee a \longrightarrow \square^m$. Now, g is injective by hypothesis. On the other hand, h is a map of representables, and E is the left-Kan extension. So $E(\mathrm{Coord})(h) = \mathrm{Coord}(h)$ is the injective monotone map $[n] \to [m]$ corresponding to h.

Finally, we show the two images are disjoint. Let $c \in \{0,1\}^m$ be the image of the glued vertex $\mathrm{fin}(\bigvee a) = \mathrm{init}(\square^n)$. Maps into a cube are monotone, raising one coordinate from 0 to 1 per edge. Since h starts at c and only raises coordinates, every direction free in h is 0 at c; since g ends at c, every direction g raises has already reached 1 at c. Hence

$$\operatorname{im} E(\mathrm{Coord})(h) \subseteq \{k : c_k = 0\} \quad \text{and} \quad \operatorname{im} E(\mathrm{Coord})(g) \subseteq \{k : c_k = 1\}$$

are disjoint. So $E(\mathrm{Coord})(f) = E(\mathrm{Coord})(g) \sqcup E(\mathrm{Coord})(h)$ is the disjoint union of injective functions with disjoint images. Therefore $E(\mathrm{Coord})(f)$ is injective. □

Lemma 3.5. *For any $f : \bigvee a \longrightarrow \bigvee b$ of morphisms in $\square_*^*$, the function $coordMap(f)$ is bijective.*

Proof. First, let's do injectivity. To begin with, we induct on the length of b. If b is empty, then $\bigvee b = \square^0$, and $\bigvee a$ only has 0-cells. Which means a must also be empty, and $f : \square^0 \to \square^0$. And $coordMap(f) : [0] \to [0]$ is always injective.

Now, consider $f : \bigvee a \longrightarrow \square^m \vee \bigvee b$, such that for any x, $\bigvee x \to \bigvee b$ is injective. By strong monoidality of Ch, f as a morphism of $Ch(\square^m \vee \bigvee b)$ splits into (f_1, f_2) as a morphism in $\mathrm{Ch}(\square^m) \times \mathrm{Ch}(\bigvee b)$. Now, by the induction hypothesis, f_2 is injective and f_1 is injective by the lemma above. Then, $f \simeq f_1 \vee f_2$, and $E(\mathrm{Coord})$ is strong monoidal. So

$$E(\mathrm{Coord})(f) = E(\mathrm{Coord})(f_1 \vee f_2) = E(\mathrm{Coord})(f_1) \sqcup E(\mathrm{Coord})(f_2)$$

which is the disjoint union of functions with disjoint codomains. Therefore f is injective.

For surjectivity, we take a counting argument. The altitudes of $\bigvee a \longrightarrow \bigvee b$ must be equal since the map is bipointed. Futhermore, the altitude of $\bigvee a = \sum_i a_i$, so $\sum_i a_i = \sum_i b_i$. So $coordMap(f)$ is a cardinality-preserving injection, hence bijective. □

4 The Complexification Construction

The construction starts by observing that one dimensional serial wedges behave like topes, or chambers, from the hyperplane configuration world. That is, they are the maximally refined elements. Let's start by reviewing the story on the braid arrangement side.

Following Paris, we specialize the story to the braid arrangement. The braid arrangement A_{n-1} is the family of hyperplanes

$$H_{ij} = \{x \in \mathbb{R}^n : x_i = x_j\}, \qquad 1 \leq i < j \leq n,$$

one for each pair of coordinates. Orient H_{ij} by the sign of $x_i - x_j$. Each point $x \in \mathbb{R}^n$ then assigns to every hyperplane a sign in $\{-, 0, +\}$, and the sign vectors so obtained are the **covectors**, or faces, of the arrangement:

$$\mathbf{Face}(n) := \{\, F : \{(i,j) : 1 \leq i < j \leq n\} \to \{-, 0, +\} \;\mid\; F \text{ arises from some } x \in \mathbb{R}^n \,\}.$$

A covector records the weak order of the coordinates of x, so it is the same data as an *ordered set partition* of $\{1,\ldots,n\}$: the blocks are the coordinates of equal value, listed in increasing order of that value. We order covectors by

$$F \le G \quad :\Longleftrightarrow \quad \forall h,\ F(h) = 0 \ \vee\ F(h) = G(h),$$

so that G resolves some of the ties left open by F. The maximal covectors — those never taking the value 0 — are the **topes**. A tope has all coordinates distinct, hence is a total order on $\{1,\ldots,n\}$; there are exactly $n!$ of them, one per permutation.

Over $\mathbb{R}$ this poset is combinatorially inert: it has a least element (the all-0 covector, the diagonal $x_1 = \cdots = x_n$), so its order complex is contractible. Salvetti's construction reruns the story over $\mathbb{C}$ and recovers the homotopy type of the complexified complement. It is captured by the **Salvetti poset**

$$\mathrm{Sal}(n) := \{\,(F,C)\ \mid\ F \in \mathbf{Face}(n),\ C \text{ a tope},\ F \le C\,\},$$

whose order uses the composition of covectors,

$$(F \circ G)(h) := \begin{cases} F(h) & \text{if } F(h) \neq 0, \\ G(h) & \text{if } F(h) = 0, \end{cases}$$

namely

$$(F,C) \preceq (F',C') \quad :\Longleftrightarrow \quad F \le F' \ \wedge\ C' = F' \circ C.$$

By Salvetti's theorem $|\,\mathrm{Sal}(n)|$ is homotopy equivalent to the complement of the complexified braid arrangement in $\mathbb{C}^n$ — the ordered configuration space of n points in $\mathbb{C}$ — so $\pi_1|\,\mathrm{Sal}(n)|$ is the pure braid group P_n. Intuitively, each face is annotated with a compatible tope, a local orientation; crossing a face can flip the orientation, and these flips are the braid letters. The 'Configuration Spaces' construction retains the same orientation data, breaking every tie at a face in all possible ways.

This packages as a Grothendieck construction, which is what we will imitate. Let

$$\mathit{Tope} : \mathbf{Face}(n) \Rightarrow \mathit{Set}, \qquad \mathit{Tope}(F) := \{C \text{ a tope} : F \le C\},$$

be the covariant functor sending $F \le F'$ to the map $C \mapsto F' \circ C$ (composing on the left by F' carries a tope above F to one above F'). Then

$$\mathrm{Sal}(n)\ \cong\ \textstyle\int \mathit{Tope},$$

the Grothendieck construction: its objects are the pairs (F,C) with $C \in \mathit{Tope}(F)$, and its morphisms are exactly the relations $\preceq$ above.

To rebuild this generically on the precubical side, let's start with the corresponding definition of tope.

Definition 4.1. A serial wedge is **one dimensional** if all its entries are 1

Definition 4.2. Let $\mathrm{Run}(K)$ be the full subcategory of $\mathrm{Ch}(K)$ such that the domains are one dimensional serial wedges.

Lemma 4.3. *Being one dimensional is preserved under* $\vee$*, so* Run *is a monoidal functor. And in paritcular* $\mathit{List}\mathbb{N} \Rightarrow \mathrm{Run}(\bigvee \cdot)$ *is a lax monoidal functor.*

Now, the morphisms we want to build are a bit unusual, because they are built from degenerecies, which are not morphisms in Precubical. That is, we want to build a map from an op arrow in $\mathrm{Ch}(K)$ to runs that mirrors the order from Salvetti.

$$(\bigvee a \longrightarrow \bigvee b) \to (\mathrm{Run}(\bigvee b) \to \mathrm{Run}(\bigvee a))$$

To achieve this, let $\mathrm{Edge}(K)$ be the chains in $\mathrm{Refine}(K)$ with each cell 1 dimensional. Then $\mathrm{Run}(K) = \mathrm{Edge}(K)$.

Now, given a face embedding $(f : \square^n \longrightarrow \square^m)$, it defines a unique projection of cells of m to cells of n, non-increasing on dimension. Call it $\hat{f}$. So we define

$$\rho : (f : \square^n \longrightarrow \square^m) \to (\mathrm{Run}[\square^m] \to (\mathrm{Run}[\square^n]))$$

via

$$\rho' : (f : \square^n \longrightarrow \square^m) \to (\mathrm{Refine}_1[\square^m] \to (\mathrm{Refine}_1[\square^n]))$$

by

$$[c_1, ..., c_k] \mapsto \text{normalize } [\hat{f}(c_1), ..., \hat{f}(c_k)]$$

where normalize removes all the 0-cells. Note that this is indeed a one dimensional chain in $\square^n$ because

1. $\hat{f}$ never increases dimension, so all dimensions are less than or equal to 1
2. normalization removes all the 0 cells, so very cell is one dimensional
3. every projection respects initial/final points

Lemma 4.4. $\rho : \mathrm{Cube}^{op} \Rightarrow Set$ *is a functor, and* $\rho(\square^0)$ *is a singleton*

Proof. The proof of functorality mostly just exploits the fact that cubes are NSL and admit altitudes. So $\mathrm{Refine}_1(\square^n) \simeq \mathrm{Run}(\square^n)$ as sets. First, ρ preserves identities, as the projection for with the identity embedding is also the identity. Second, if $f : \square^n \to \square^m$ and $g : \square^m \to \square^n$ are face embeddings, then their corresponding projections respect composition.

For the singleton, observe that there $\rho(\square^0) = \mathrm{Run}[\square^0]$, but the only one dimensional serial wedge which maps into $\square^0$ is the the empty one. So indeed $\rho(\square^0)$ is a singleton. $\square$

That means two things. Firstly, it means ρ is itself, a precubical set. Second, it means our lifting machinery sends ρ to $E(\rho) : \square_*^{*op} \Rightarrow Cat$. Lastly, let $\square^\vee$ be the full subcategory of serial wedges in $\square_*^*$. There is also a forgetful functor $W(K) : \mathrm{Ch}(K) \Rightarrow \square^\vee$ sending $\bigvee a \to K \mapsto \bigvee a$. Then we can define our key functor

$$\mathrm{Lines}(K) := E(\rho) \circ W(K)^{op}$$

which is a presheaf on $\mathrm{Ch}(K)$. Then we define

$$\mathrm{Ch}^*(K) := \int \mathrm{Lines}$$

the category of elements of the presheaf $\mathrm{Lines}(K)$. Defining Ch^* abstractly has some convienences later. Let's unpack the definition of Ch^* to have a more concrete presentation of it.

Objects of $\mathrm{Ch}^*(K)$ are pairs

$$(\bigvee a \longrightarrow K, x : \mathrm{Run}[\bigvee a]), \text{ I.E. } \bigvee[1, ..., 1] \longrightarrow \bigvee a \longrightarrow K$$

The intuition here matches the Salvetti story precisely, we have a wedge map into K, coupled with orientation info about it. Morphisms are a little uglier. A morphism

$$(\bigvee a \longrightarrow K, x : \mathrm{Run}[\bigvee a]) \longrightarrow (\bigvee b \longrightarrow K, y : \mathrm{Run}[\bigvee b])$$

unpacks to

1. A morphism $f : (\bigvee a \longrightarrow K) \longrightarrow (\bigvee b \longrightarrow K)$ I.E. a commuting $f' : \bigvee a \longrightarrow \bigvee b$.
2. $x = E(\rho)(f')(y)$, equivalently $\phi : x \to \mathrm{Lines}(f')(y)$: we take the run y, project it along f', and require the result to be x. Since the fibers of $E(\rho)$ are discrete, this is a condition, not extra data — x is determined by f' and y.

We'd like to associate a permutation with each such morphism. Note that each run $r \in \mathrm{Run}[\bigvee p]$ is a bipointed map $\bigvee[1, \dots, 1] \to \bigvee p$, so by bijectivity of $coordMap$ it is the same as a linear ordering of the coordinates of $\bigvee p$:

$$\bar{r} := coordMap(r) : [n] \xrightarrow{\sim} \textstyle\coprod_i [p_i].$$

Conjugating the coordinate bijection $coordMap(f') : \coprod_i [a_i] \to \coprod_j [b_j]$ by the two run orderings gives a permutation of $[n]$,

$$S(f) := \bar{y}^{-1} \circ coordMap(f') \circ \bar{x} \;\in\; \Sigma_n.$$

This permutation $S(f)$ is what feeds the braid functor.

Conjugation is what makes it functorial: along a composite the middle ordering $\bar{y}$ cancels,

$$S(g \circ f) = \bar{z}^{-1}\, coordMap(g)\, coordMap(f)\, \bar{x} = (\bar{z}^{-1} coordMap(g)\, \bar{y})\,(\bar{y}^{-1} coordMap(f)\, \bar{x}) = S(g)\, S(f),$$

and

$$S(\mathrm{id}) = \bar{z}^{-1}\, coordMap(\mathrm{id}) \bar{z} = \mathrm{id}$$

so

Lemma 4.5. $S : \mathrm{Ch}^*(K) \to \bigsqcup_n \Sigma_n$ *is a functor.*

The diagram below records the story. The solid arrows are morphisms of $\mathrm{Ch}^*(K)$; the dashed $S(f)$ (a permutation of positions) and $E(\rho)(f)$ (the run projection) are plain functions, not morphisms in $\square^*_*$, and they point oppositely — $E(\rho)(f)$ carries y back to x, and $S(f)$ is the induced reindexing.

$$\begin{tikzcd}
[1, \dots, 1] \arrow[r, "x"] \arrow[d, dashed, "S(f)"'] & \bigvee a \arrow[d, "f"] \arrow[dr] & \\
[1, \dots, 1] \arrow[r, "y"'] & \bigvee b \arrow[r] \arrow[u, dashed, bend left, "E(\rho)(f)"] & K
\end{tikzcd}$$

5 The Braiding

You'll notice that the K side is not involved in the computation of the permutation, or the proofs of its functoriality.

This comes from a rather nice fact that about grothendieck constructions:

$$\begin{tikzcd}
\mathrm{Ch}^*(K) = \int(E(\rho)\circ W(K)) \arrow[r, "Q"] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner", very near start] & \int E(\rho) = \square^\vee/\rho \arrow[d, "\pi"] \\
\mathrm{Ch}(K) \arrow[r, "W(K)"'] & \square^\vee
\end{tikzcd}$$

The map Q is just a projection to the first arrow of $[1..., 1] \to \bigvee a \to K$. And ρ is the classifier for runs. Since E is essentialy just the yoneda embedding, the right corner turns into a slice category of wedges to ρ. Secondly, K does not appear on the right hand side.

Lemma 5.1. *The functor Q is always faithful.*

Proof. Let x, y be two objects in $\mathrm{Ch}^*(K)$. Then let $[1, ..., 1] \to \bigvee a \to K$ and $[1, ..., 1] \to \bigvee b \to K$ be two objects in $\mathrm{Ch}^*(K)$. Two morphisms f, g between these objects are equal exactly when they have the same map $\bigvee a \to \bigvee b$ Which happens if and only if $Q(f) = Q(g)$. So Q is indeed faithful. $\square$

But, we can do a nicer trick

Lemma 5.2. *Let Z be the final precubical set. Then $\mathrm{Ch}^*(Z)$ is isomorphic to $\square^\vee/\rho$.*

Proof. We know Q is faithful already. To show Q is full, again we have two objects $[1, ..., 1] \to \bigvee a \to K$ and $[1, ..., 1] \to \bigvee b \to K$. Let f be a morphism in $\square^*_*/\rho$. That is, $f : \bigvee a \to \bigvee b$ and it forms a commuting triangle:

$$\begin{tikzcd}
\bigvee a \arrow[r] \arrow[d, "f"'] & \rho \\
\bigvee b \arrow[ur] &
\end{tikzcd}$$

But, since Z is the final precubical set, every map commutes in the triangle

$$\begin{tikzcd}
\bigvee a \arrow[r] \arrow[d, "f"'] & Z \\
\bigvee b \arrow[ur] &
\end{tikzcd}$$

So Q is full.

Surjectivity on objects is equally straightforward. Every wedge $\bigvee a \to \rho$ is equivalent to a $[1, ...1] \to \bigvee a$. And every $\bigvee a$ has a unique $\bigvee a \to Z$. So we have a corresponding object of $\mathrm{Ch}^*(Z)$. $\square$

But we need a device for building braid words from permutations. Which brings us to the Garside Germ Presentation of braids.

That is, the set $\{[\sigma] \mid \sigma \in \Sigma_n \wedge [\sigma\sigma'] = [\sigma] + [\sigma'] \text{ when } |\sigma\sigma'| = |\sigma||\sigma'|\}$ where

$$i(\sigma) := \{(i, j) \in [n] \times [n] | i < j \wedge \sigma(j) < \sigma(i)\}$$

and $|\sigma| = |i(\sigma)|$. This is the set inversions. The key fact is a "no double crossings" feature of Ch^*. That is, once you refine a coordinate, you can't refine it again. Let's first state this in pure combinatorics

Lemma 5.3. *let f and g be permutations of $[n]$. And suppose that $i(f) \subset i(g \circ f)$. Then $|g \circ f| = |g| + |f|$.*

Proof. Every element of $i(g \circ f)$ either lies in $i(f)$ or not, so

$$i(g \circ f) \;=\; \big(i(g \circ f) \cap i(f)\big) \;\sqcup\; \big(i(g \circ f) \setminus i(f)\big).$$

By hypothesis $i(f) \subseteq i(g \circ f)$, so the first set equals $i(f)$ and contributes $|f|$.

For the second, a pair $(k, l) \notin i(f)$ has $f(k) < f(l)$, and then $(k, l) \in i(g \circ f)$ iff $g(f(k)) > g(f(l))$, i.e. iff $(f(k), f(l)) \in i(g)$. Since f is a bijection, $(k, l) \mapsto (f(k), f(l))$ restricts to a bijection

$$i(g \circ f) \setminus i(f) \;\xrightarrow{\sim}\; \{(p, q) \in i(g) \;:\; f^{-1}(p) < f^{-1}(q)\}.$$

This target is all of $i(g)$: otherwise take $(p, q) \in i(g)$ and set $k = f^{-1}(q) < l = f^{-1}(p)$; then $f(k) = q > p = f(l)$ gives $(k, l) \in i(f)$, while $g(f(k)) = g(q) < g(p) = g(f(l))$ gives $(k, l) \notin i(g \circ f)$, contradicting $i(f) \subseteq i(g \circ f)$. Hence the second set contributes $|g|$, and $|g \circ f| = |f| + |g|$. $\square$

Lemma 5.4. *The map $f \mapsto [S(f)]$ gives us a functor* $\mathrm{Ch}^*(Z) \Rightarrow$ *Braid, which lifts to* Conc $:\Rightarrow$ *Braid*

Proof. It always sends objects to the braid object associated with the length of x.

For morphisms, we need to show is that $|S(f \circ g)| = |S(f)| + |S(g)|$ in order for this to be defined.

Let $g : \bigvee a \to \bigvee b$ and $f : \bigvee b \to \bigvee c$. According to the previous lemma, it suffices to show that $i(S(f)) \subset i(S(g \circ f))$ Since Ch is monoidal, we can split $\mathrm{Ch}(\bigvee c)$ into $\Pi_i(Ch(\square^{c_i}))$. This splits the triangle of $\bigvee a \to \bigvee b$ over $\bigvee c$ into one diagram per component. Since *coordMap* is also monoidal, it suffices to prove this for each diagram.

So, we can assume $f : \bigvee b \to \square^m$. Suppose $(k, l) \in i(f)$. What makes the precubical sets specials is that maps from cubes to cubes respects the order embedding. So $\bigvee b \to \square^m$ must split as $(\bigvee b_l) \vee (\bigvee b_r) \to \square^m$ such that at the junction vertex, the k coordinate is 1, but the i coordinate is 0. Now, $g : \bigvee a \to (\bigvee b_l) \vee (\bigvee b_r)$ splits through $\mathrm{Ch}((\bigvee b_l) \vee (\bigvee b_r))$. into $g_l : \bigvee a_l \to \bigvee b_l$ and $g_r : \bigvee a_r \to \bigvee b_r$. In other words, $g \circ f = (g_l \circ f_l) \vee (g_r \circ f_r)$ with the same junction point. So $(k, l) \in i(g \circ f)$.

Composition is now well-defined, and in fact $[S(g \circ f)] = [S(g)][S(f)]$. Also $[S(\mathrm{id})] = \mathrm{id}$, so indeed Conc defines a functor. $\square$

ClaudeSlop to review:

Lemma 5.5. *The map $f \mapsto [S(f)]$ is a functor* $\mathrm{Ch}^*(Z) \Rightarrow$ *Braid, which lifts to* Conc.

Proof. On objects, S sends a wedge to the braid object indexed by its length, and $[S(\mathrm{id})] = \mathrm{id}$. For composition, let $g : \bigvee a \to \bigvee b$ and $f : \bigvee b \to \bigvee c$ be composable in Ch. Since $S = \mathit{coordMap}$ is a functor we have $S(f \circ g) = S(f) \circ S(g)$, so by the previous ("no double crossing") lemma, applied with $\phi = S(g)$ and $\psi = S(f)$, it suffices to prove

$$i(S(g)) \subseteq i(S(f \circ g)). \tag{$*$}$$

Write $\bigvee b = \bigvee[b_1, \ldots, b_r]$ and identify $\mathit{coordMap}(\bigvee b) = \coprod_j [b_j]$, ordered first by summand and then within each summand.

Claim 1: every inversion of $S(g)$ has both coordinates in a common summand of $\bigvee b$. Since $coordMap(g)$ is monotone in its first (summand) coordinate, $k < l$ implies that $S(g)(k)$ and $S(g)(l)$ lie in summands $s_k \leq s_l$ of $\bigvee b$. If $s_k < s_l$, then $S(g)(k)$ lies in a strictly earlier summand than $S(g)(l)$ and hence $S(g)(k) < S(g)(l)$; but (k, l) is an inversion, so $S(g)(k) > S(g)(l)$. Therefore $s_k = s_l$: both coordinates lie in a common summand of $\bigvee b$.

Claim 2: $S(f)$ preserves order inside each summand of $\bigvee b$. By naturality of $coordMap$, its restriction to the block $[b_j]$ is $coordMap$ of the composite $\square^{b_j} \hookrightarrow \bigvee b \xrightarrow{f} \bigvee c$. This is a map of cubes into a serial wedge, so it lands in a single summand of $\bigvee c$ as a face embedding, and $coordMap$ of a face embedding is an order-preserving injection. Thus for $p < q$ in a common block $[b_j]$ we have $S(f)(p) < S(f)(q)$.

Combining. Let $(k, l) \in i(S(g))$ and set $p = S(g)(l) < q = S(g)(k)$. By Claim 1 the pair p, q lies in a common summand of $\bigvee b$, so by Claim 2, $S(f)(p) < S(f)(q)$. Hence

$$S(f \circ g)(k) = S(f)(q) > S(f)(p) = S(f \circ g)(l),$$

and since $k < l$ this is exactly $(k, l) \in i(S(f \circ g))$, proving $(*)$. Therefore $|S(f \circ g)| = |S(f)| + |S(g)|$, so $[S(f \circ g)] = [S(f)][S(g)]$ and $f \mapsto [S(f)]$ is a functor. $\square$

This looks a bit silly, because all we've done is proven our braids words have no cancellation. But that only means we found a nice basis. And in fact, it means we're working the positive braid monoid. We recover the full braiding structure by lifting to the FreeGroupoid, which is now free.

Lemma 5.6. *For any precubical set K, if there is an $f : \square^2 \longrightarrow K$, then* $\mathrm{Conc}(K)$ *is nontrivial*

Proof. It suffices to find a morphism f in $\mathrm{Ch}^*(K)$ such that $S(Q(f))$ is not the identity. Consider two maps $p : \bigvee[1, 1] \to \bigvee[2]$ going along the bottom edge, then the right edge. And consider the map $q \bigvee[1, 1] \to \bigvee[2]$ going along the top edge. We have

where $\sigma(p) = \{1 \mapsto 2, 2 \mapsto 1\}$ has an inversion. so $[\sigma(p)]$ is not the identity braid. $\square$

Lemma 5.7. *The category* $\mathrm{Conc}(\square^n)$ *is isomorphic to the pure braids.*

6 Relationship to the Salvetti Complex

The key fact here is that our story mirrors the salvetti story, on the nose, for cubes. That is, the face complex is exactly $\mathrm{Ch}(\square^n)$, and the salvetti complex is exactly $\mathrm{Ch}^*(\square^n)$. I'd argue this explains why the braid invariant from 'Configuration Spaces' collapsed, and why this complexification survives. However, it does suggest that

1. The salvetti complex is also $\mathrm{Ch}(Z)$, so the quotiont by the free action some is also related to the complexification story
2. $\mathrm{Ch}^*(Z)$ is bigger than Braid, so something interesting is happening